

## Technology Stack

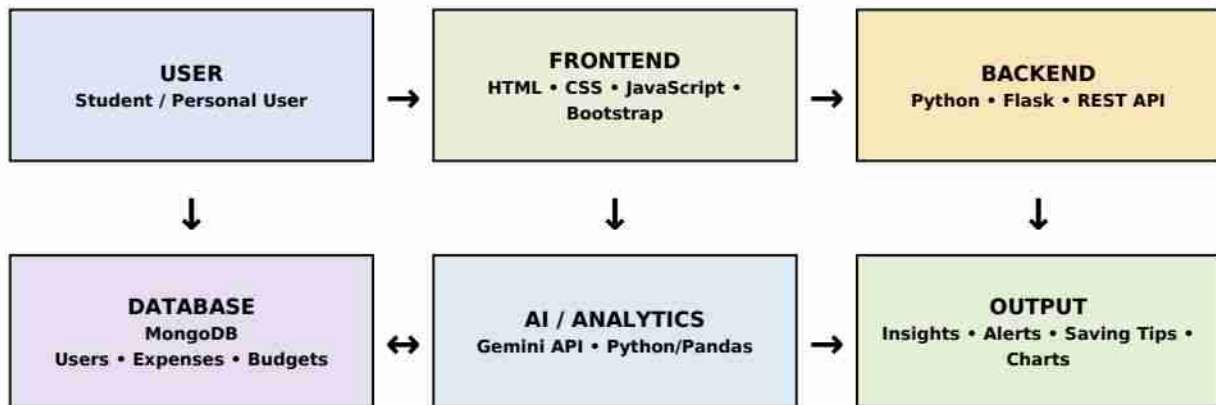
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|---------------|-------------------|
| Date          | 29 September 2026 |
| Team ID       | SWTID-2026-2087   |
| Project Name  | POCKET SMART AI   |
| Maximum Marks | 3 Marks           |

### Technology Stack - POCKET SMART AI

The technology stack defines the technologies, frameworks, APIs and development tools used to build the PocketSmart AI solution. The stack is organized from the user interface through backend processing, database storage, AI services, analytics, visualization and deployment.

| S.No | Technology Layer  | Technology / Tool                 | Purpose in PocketSmart AI                                                                                          | Role          |
|------|-------------------|-----------------------------------|--------------------------------------------------------------------------------------------------------------------|---------------|
| 1    | Frontend          | HTML5, CSS3, JavaScript           | Build the responsive user interface, forms, dashboard, expense screens and interactive components.                 | Core          |
| 2    | UI / Styling      | Bootstrap 5                       | Provide responsive layouts, cards, tables, forms and mobile-friendly components.                                   | Core          |
| 3    | Backend           | Python + Flask                    | Handle application logic, API requests, validation, authentication flow and communication between UI and services. | Core          |
| 4    | Database          | MongoDB                           | Store user profiles, expenses, income, categories, budgets and generated financial records.                        | Core          |
| 5    | AI / LLM          | Google Gemini API                 | Generate spending insights, natural-language explanations and personalized saving suggestions from processed data. | AI            |
| 6    | Data Processing   | Python / Pandas                   | Clean, organize and analyse transaction data for summaries, categories and spending patterns.                      | Analytics     |
| 7    | Charts & Reports  | Chart.js                          | Display expense breakdowns, budget progress and spending trends in the dashboard.                                  | Visualization |
| 8    | API Format        | REST API / JSON                   | Transfer structured data between the frontend, backend and AI service.                                             | Integration   |
| 9    | Development Tools | Visual Studio Code + Git + GitHub | Write, test, version-control and collaborate on the project source code.                                           | Tools         |

## POCKET SMART AI – Technology Architecture



## How the Technology Stack Works

| Step | Layer              | Data / Function Flow                                                                                                       |
|------|--------------------|----------------------------------------------------------------------------------------------------------------------------|
| 1    | User Interface     | The student enters expenses, income and budget information through the responsive web interface.                           |
| 2    | Backend Processing | Flask receives the request, validates the input and applies application/business logic.                                    |
| 3    | Database           | MongoDB stores the structured user, transaction and budget information for later retrieval.                                |
| 4    | AI & Analytics     | Python/Pandas processes transaction data and the Gemini API can generate natural-language insights and saving suggestions. |
| 5    | Visualization      | Chart.js presents spending categories, budget progress and trends in an easy-to-understand format.                         |
| 6    | User Output        | The dashboard returns useful summaries, alerts and recommendations to help the user make informed budgeting decisions.     |

Final Technology Flow: User → Frontend → Flask Backend → MongoDB + AI/Analytics → Charts, Alerts & Personalized Insights → User Dashboard.